

Sequential Optimization Architecture

minimize $f(\mathcal{A})$ **s.t.** $\mathcal{A} \in \mathcal{X}$
(over $\mathbf{x}_g$)

update $\mathbf{x}_g$

minimize $f_1(\mathcal{A})$ **s.t.** $[\mathbf{x}^{(1)}, \mathbf{y}^{(1)}] \in \mathcal{X}_1$

update $\mathbf{x}_i^{(1)}$

update $\mathbf{y}$

**Disciplinary
Analysis 1**

minimize $f_2(\mathcal{A})$ **s.t.** $[\mathbf{x}^{(2)}, \mathbf{y}^{(2)}] \in \mathcal{X}_2$

update $\mathbf{x}_i^{(2)}$

update $\mathbf{y}$

**Disciplinary
Analysis 2**

$\vdots$

minimize $f_m(\mathcal{A})$ **s.t.** $[\mathbf{x}^{(m)}, \mathbf{y}^{(m)}] \in \mathcal{X}_m$

**Disciplinary
Analysis m**